Energy Saving Agent

An Intelligent Agent Design Based on the PEAS Framework

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1. Introduction

Energy wastage is a common problem in houses, factories, and other places in Pakistan. Most of the time it happens because of small habits, like leaving a fan on after leaving the room, or letting the AC run even after the room has already cooled down. This report explains an Energy Saving Agent, a simple system that keeps an eye on the room and turns things off automatically when they are not needed.

2. Problem Statement

Energy gets wasted in homes, factories, and offices mostly because no one is watching the appliances all the time. For example, someone turns on a fan, leaves the room, and forgets to turn it off, so the fan keeps running for no reason. In the same way, an AC keeps cooling a room even after it has already reached a comfortable temperature, which also wastes energy. These small things don't look like much on their own, but they add up to a lot of wasted energy and money over time.

3. Proposed Solution

To fix this, I designed an Energy Saving Agent that keeps track of energy usage in homes, industries, and other places. It checks if the room is occupied, checks the temperature, and checks if any lights are on for no reason. Based on what it senses, it automatically turns things on or off, so no one has to do it manually.

4. Decision Rules

The agent basically follows a few simple rules:

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IF room is empty
   → Lights OFF
   → Fan OFF
   → AC OFF

IF room is occupied AND temperature > 30°C
   → AC ON

IF room is occupied AND temperature ≤ 30°C
   → AC OFF
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5. PEAS Framework

We can describe this agent using the PEAS framework (Performance, Environment, Actuators, Sensors), explained below.

Performance Measure The agent is doing well if it catches energy wastage and reacts correctly. Right now it's basically 100% accurate for the situations it's built to handle.	**Environment** The environment is the room the agent is placed in.
Actuators This is how the agent actually does something. It turns the lights, fan, and AC on or off depending on what it senses.	**Sensors** This is how the agent "sees" the room. It checks the temperature, whether lights are on unnecessarily, and whether the room is occupied.

Environment Types:

- Observability: Fully Observable (type 1).
- Determinism: Deterministic (type 1).
- Episodes: Episodic (type 1).
- Change: Static (type 1).
- Values: Discrete (type 1).
- Agents: Single Agent (type 1).

6. Agent-Environment Interaction Diagram

Figure 1 below shows how this works in a simple way. The sensors pick up info from the room, pass it to the agent, and the agent tells the actuators what to do.

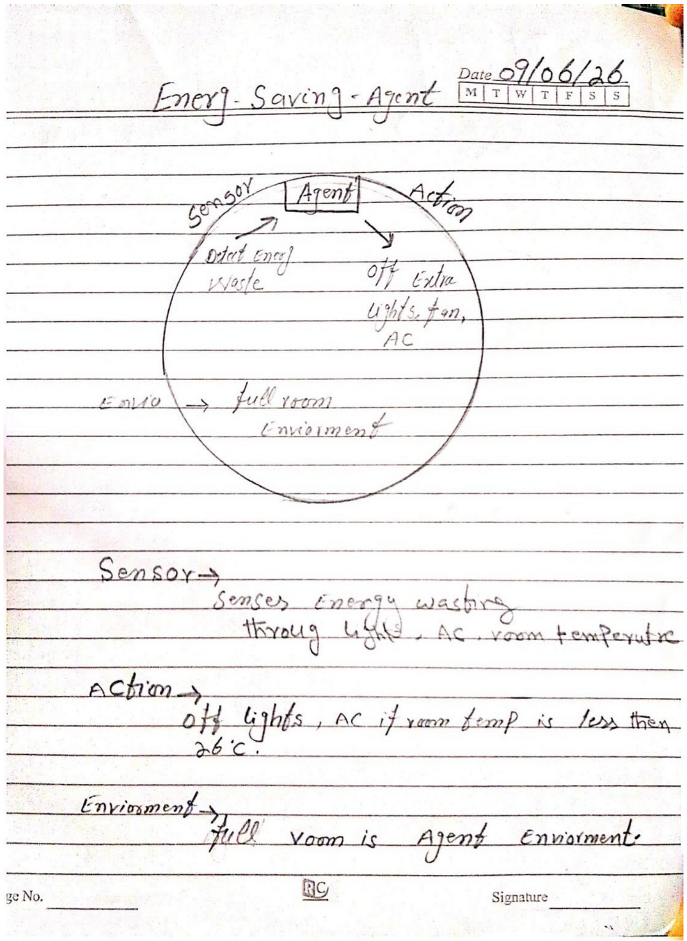

Figure 1: Agent-Environment Interaction Diagram

7. Conclusion

Overall, this Energy Saving Agent shows that even a simple set of rules can cut down a lot of energy waste in everyday places. Since it keeps checking occupancy, temperature, and lights on its own, no one needs to remember to turn things off. The agent handles it, so energy only gets used when it's actually needed.